

Naive Bayes classifier

A Naive Bayes classifier is a probabilistic machine learning model that's used for classification task. The crux of the classifier is based on the Bayes theorem.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Using Bayes theorem, we can find the probability of A happening, given that B has occurred. Here, B is the evidence and A is the hypothesis.

The assumption made here is that the predictors/features are independent. That is presence of one particular feature does not affect the other. Hence it is called naive.

Where,

P(A|B) is Posterior probability: Probability of hypothesis A on the observed event B.

P(B|A) is Likelihood probability: Probability of the evidence given that the probability of a hypothesis is true.

P(A) is Prior Probability: Probability of hypothesis before observing the evidence.

P(B) is Marginal Probability: Probability of Evidence.

If we consider Bayes Theorem in another representation,

$$P(y|X) = \frac{P(X|y)P(y)}{P(X)}$$

The variable y is the class variable, Variable X represent the parameters/features.

Bayesian equation to calculate the posterior probability for each class. The class with the highest posterior probability is the outcome of the prediction.

Example for applying Bayes theorem on data set with single feature:

here add---

Suppose we have a dataset of weather conditions and corresponding target variable "Play". So using this dataset we need to decide that whether we should play or not on a particular day according to the weather conditions.

Problem: If the weather is sunny, then the Player should play or not?

	Outlook	Play
0	Rainy	Yes
1	Sunny	Yes
2	Overcast	Yes
3	Overcast	Yes
4	Sunny	No
5	Rainy	Yes
6	Sunny	Yes
7	Overcast	Yes
8	Rainy	No
9	Sunny	No
10	Sunny	Yes
11	Rainy	No
12	Overcast	Yes
13	Overcast	Yes

Frequency table for the Weather Conditions:

Weather	Yes	No
Overcast	5	0
Rainy	2	2
Sunny	3	2
Total	10	5

Probabilities/likelihood table:

Weather	No	Yes	
Overcast	0	5	$5/14 = 0.35$
Rainy	2	2	$4/14 = 0.29$
Sunny	2	3	$5/14 = 0.35$
All	$4/14 = 0.29$	$10/14 = 0.71$	

Applying Bayes' theorem:

$$P(\text{Sunny} | \text{Yes}) = 3/10 = 0.3$$

$$P(\text{Sunny} | \text{NO}) = 2/4 = 0.5$$

$$P(\text{Sunny}) = 0.35$$

$$P(\text{Sunny}) = 0.35$$

$$P(\text{Yes}) = 0.71$$

$$P(\text{No}) = 0.29$$

$$P(\text{Yes} | \text{Sunny}) = 0.3 * 0.71 / 0.35 = 0.60$$

$$P(\text{No} | \text{Sunny}) = 0.5 * 0.29 / 0.35 = 0.41$$

Therefore $P(\text{Yes} | \text{Sunny}) > P(\text{No} | \text{Sunny})$

Hence on a Sunny day, Player will play the game.

Let us now consider data set with multiple features in a feature /attribute set X:

$$X = (x_1, x_2, x_3, \dots, x_n)$$

By substituting for X, we get,

$$P(y|x_1, \dots, x_n) = \frac{P(x_1|y)P(x_2|y)\dots P(x_n|y)P(y)}{P(x_1)P(x_2)\dots P(x_n)}$$

For all entries in the dataset, the denominator does not change, it remain static.

Therefore, the denominator can be removed and a proportionality can be introduced.

$$P(y|x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i|y)$$

We need to find the class **y** with maximum probability.

$$y = \operatorname{argmax}_y P(y) \prod_{i=1}^n P(x_i|y)$$

The posterior probability $P(y|X)$ can be calculated by first, creating a Frequency Table for each attribute against the target. Then, molding the frequency tables to Likelihood Tables and finally, use the Naïve Bayesian equation to calculate the posterior probability for each class. The class with the highest posterior probability is the outcome of the prediction.

Example2:

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

Concerning our dataset, the concept of assumptions made by the algorithm can be understood as:

- We assume that no pair of features are dependent. For example, the color being 'Red' has nothing to do with the Type or the Origin of the car. Hence, the features are assumed to be Independent.

Secondly, each feature is given the same influence(or importance). For example, knowing the only Color and Type alone can't predict the outcome perfectly. So none of the attributes are irrelevant and assumed to be contributing Equally to the outcome.

Here in our dataset, we need to classify whether the car is stolen, given the features of the car.

Frequency Table

		Stolen?	
		Yes	No
Color	Red	3	2
	Yellow	2	3



Likelihood Table

		Stolen?	
		P(Yes)	P(No)
Color	Red	$3/5$	$2/5$
	Yellow	$2/5$	$3/5$

Frequency and Likelihood tables of 'Color'

Frequency Table

		Stolen?	
		Yes	No
Type	Sports	4	2
	SUV	1	3



Likelihood Table

		Stolen?	
		P(Yes)	P(No)
Type	Sports	$4/5$	$2/5$
	SUV	$1/5$	$3/5$

Frequency and Likelihood tables of 'Type'

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Frequency Table

		Stolen?	
		Yes	No
Origin	Domestic	2	3
	Imported	3	2



Likelihood Table

		Stolen?	
		P(Yes)	P(No)
Origin	Domestic	$2/5$	$3/5$
	Imported	$3/5$	$2/5$

Frequency and Likelihood tables of 'Origin'

Given test record:

Color	Type	Origin	Stolen
Red	SUV	Domestic	?

Applying Naïve Bayes classifier:

$$\begin{aligned}P(\text{Yes} \mid X) &= P(\text{Red} \mid \text{Yes}) * P(\text{SUV} \mid \text{Yes}) * P(\text{Domestic} \mid \text{Yes}) * P(\text{Yes}) \\&= \frac{3}{5} * \frac{1}{5} * \frac{2}{5} * 1 \\&= 0.048\end{aligned}$$

$$\begin{aligned}P(\text{No} \mid X) &= P(\text{Red} \mid \text{No}) * P(\text{SUV} \mid \text{No}) * P(\text{Domestic} \mid \text{No}) * P(\text{No}) \\&= \frac{2}{5} * \frac{3}{5} * \frac{3}{5} * 1 \\&= 0.144\end{aligned}$$

Since $0.144 > 0.048$, Which means given the features RED SUV and Domestic, our example gets classified as 'NO' the car is not stolen.